

ELE2038 Control System Design

Ball-on-Incline System: Modelling, Linearisation, and PID Control

Dara McCollam, Joseph McNeill, Ciaran Coghlan

GitHub Link

1 System Description

The system consists of a wooden ball of mass m and radius r rolling without slipping on an inclined plane at angle ϕ to the horizontal. The ball is connected to a spring of stiffness k (natural length at $x = d$) and a viscous damper with coefficient b . An electromagnet located at $x = \delta > d$ exerts an attractive force on a metal core at the centre of the ball. The sign convention adopted is:

- positive direction: down the slope,
- negative direction: up the slope.

The magnetic force is

$$F_{\text{mag}} = c \frac{i^2}{y^2}, \quad (1)$$

where i is the circuit current and $y = \delta - x$ is the distance from the ball centre to the electromagnet centre (assuming $x < \delta$).

2 Equations of Motion

2.1 Force Components Along the Slope

The forces acting on the ball along the x -axis are:

1. **Gravity:** $mg \sin \phi$ (positive, acts down the slope),
2. **Spring:** $F_s = -k(x - d)$ (negative, acts up the slope when $x > d$),
3. **Damper:** $F_d = -b\dot{x}$ (opposes motion),
4. **Magnetic force** (acts towards magnet at $x = \delta$):

$$F_{\text{mag}} = \frac{ci^2}{(\delta - x)^2}, \quad (2)$$

5. **Friction:** $-T$ (rolling constraint reaction).

2.2 Newton's Second Law — Translational

Applying $\sum F_x = m\ddot{x}$:

$$m\ddot{x} = mg \sin \phi - k(x - d) - b\dot{x} + \frac{ci^2}{(\delta - x)^2} - T. \quad (3)$$

2.3 Rotational Dynamics

Since only friction T acts at a point at radius r from the centre, it is the sole torque-producing force. Applying the rotational form of Newton's second law, $\sum \tau = I\ddot{\theta}$:

$$I\ddot{\theta} = -Tr, \quad (4)$$

where the negative sign reflects that friction produces a clockwise torque (opposing rolling down the slope).

Using the rolling-without-slipping constraint $x = -r\theta$ (so that positive x corresponds to motion down the slope):

$$\dot{x} = -r\dot{\theta}, \quad (5)$$

$$\ddot{x} = -r\ddot{\theta} \implies \ddot{\theta} = -\frac{\ddot{x}}{r}. \quad (6)$$

Substituting into $I\ddot{\theta} = -Tr$:

$$\begin{aligned} I\left(-\frac{\ddot{x}}{r}\right) &= -Tr, \\ -\frac{I\ddot{x}}{r} &= -Tr, \\ T &= \frac{I}{r^2}\ddot{x}. \end{aligned} \quad (7)$$

2.4 Combined Equation of Motion

Substituting (7) into (3):

$$\begin{aligned} m\ddot{x} &= mg \sin \phi - k(x - d) - b\dot{x} + \frac{ci^2}{(\delta - x)^2} - \frac{I}{r^2}\ddot{x}, \\ \left(m + \frac{I}{r^2}\right)\ddot{x} &= mg \sin \phi - k(x - d) - b\dot{x} + \frac{ci^2}{(\delta - x)^2}. \end{aligned} \quad (8)$$

2.5 Moment of Inertia for a Solid Sphere

The ball is modelled as a solid sphere, so $I = \frac{2}{5}mr^2$. Therefore:

$$m + \frac{I}{r^2} = m + \frac{\frac{2}{5}mr^2}{r^2} = m + \frac{2}{5}m = \frac{7}{5}m. \quad (9)$$

Equation (8) becomes:

$$\frac{7}{5}m\ddot{x} = mg \sin \phi - k(x - d) - b\dot{x} + \frac{ci^2}{(\delta - x)^2}, \quad (10)$$

or equivalently,

$$\ddot{x} = \frac{5}{7m} \left(mg \sin \phi - k(x - d) - b\dot{x} + \frac{ci^2}{(\delta - x)^2} \right). \quad (11)$$

3 State-Space Representation

The nonlinear equations of motion are expressed in state-space form [1]. Define the state vector $\mathbf{x} = [x_1, x_2]^\top$ with $x_1 = x$ (position) and $x_2 = \dot{x}$ (velocity). The state equations are:

$$\dot{x}_1 = x_2, \quad (12)$$

$$\dot{x}_2 = \frac{5}{7m} \left(mg \sin \phi - k(x_1 - d) - bx_2 + \frac{ci^2}{(\delta - x_1)^2} \right). \quad (13)$$

The current i acts as the system input.

4 Equilibrium Analysis

4.1 Equilibrium Conditions

The operating point is specified as $x_0 \approx 0.50$ m. At equilibrium the velocity and acceleration are both zero:

$$v_0 = 0, \quad \dot{v}_0 = 0. \quad (14)$$

Setting $\dot{x}_2 = 0$ in (13) and noting that the $\frac{5}{7m}$ scaling factor and the $bv_0 = 0$ term both vanish:

$$mg \sin \phi - k(x_0 - d) + \frac{ci_0^2}{(\delta - x_0)^2} = 0. \quad (15)$$

4.2 Solving for the Equilibrium Current

Rearranging (15):

$$\begin{aligned} \frac{ci_0^2}{(\delta - x_0)^2} &= k(x_0 - d) - mg \sin \phi, \\ ci_0^2 &= [k(x_0 - d) - mg \sin \phi](\delta - x_0)^2, \\ i_0^2 &= \frac{[k(x_0 - d) - mg \sin \phi](\delta - x_0)^2}{c}. \end{aligned} \quad (16)$$

Therefore:

$$i_0 = \sqrt{\frac{[k(x_0 - d) - mg \sin \phi](\delta - x_0)^2}{c}}. \quad (17)$$

4.3 Existence Condition

Since $F_{\text{mag}} = ci^2/(\delta - x)^2 \geq 0$ always, a real equilibrium current exists if and only if the expression inside the square root in (17) is non-negative:

$$k(x_0 - d) - mg \sin \phi \geq 0 \iff x_0 \geq d + \frac{mg \sin \phi}{k}. \quad (18)$$

4.4 Numerical Evaluation

Using the system parameters ($m = 0.462$ kg, $g = 9.81$ m/s², $d = 0.42$ m, $\delta = 0.65$ m, $k = 1885$ N/m, $\phi = 41^\circ$, $c = 6.811 \times 10^{-3}$ kg m³ A⁻² s⁻², $b = 10.4$ N s m⁻¹):

At $x_0 = 0.40$ m:

$$i_0^2 \approx -373.23 < 0 \implies \text{no real } i_0. \quad (19)$$

This confirms that $x_0 = 0.40$ m is not a viable set-point: the spring force does not yet exceed gravity along the slope, so no attractive magnetic force can hold the ball in equilibrium.

The existence condition (17) gives the feasibility bound:

$$x_0 \geq d + \frac{mg \sin \phi}{k} = 0.42 + \frac{0.462 \times 9.81 \times \sin 41^\circ}{1885} \approx 0.4216 \text{ m}. \quad (20)$$

The feasible operating range is therefore $0.4216 \text{ m} \leq x_0 < \delta = 0.65$ m, where the upper bound $x_0 < \delta$ is required to maintain a positive gap between the ball and the electromagnet.

At $x_0 = 0.50$ m (*the selected operating point*):

Substituting into (17) with $\delta - x_0 = 0.15$ m:

$$i_0 = \sqrt{\frac{[1885(0.08) - 0.462 \times 9.81 \times \sin 41^\circ](0.15)^2}{6.811 \times 10^{-3}}} \approx 22.099 \text{ A}, \quad (21)$$

$$V_0 = Ri_0 \approx 2200 \times 22.099 \approx 4.86 \times 10^4 \text{ V}. \quad (22)$$

The linearised state-space matrices evaluated at this operating point are:

$$A = \begin{bmatrix} 0 & 1 \\ 132.995 & -16.079 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 20.685 \end{bmatrix}, \quad (23)$$

$$C = [1 \quad 0], \quad D = 0. \quad (24)$$

The positive entry $a_{21} = 132.995 > 0$ confirms the destabilising effect of the magnetic force gradient at this operating point, and the system is therefore linearised here for all subsequent controller design.

5 Linearisation

5.1 Identifying the Nonlinear Term

The only nonlinear term in (11) is the magnetic force. Define:

$$\varphi(x, i) = \frac{ci^2}{(\delta - x)^2} = ci^2(\delta - x)^{-2}. \quad (25)$$

5.2 Gradient of φ

Partial derivative with respect to x :

Using the chain rule with $\frac{d}{dx}(\delta - x) = -1$:

$$\begin{aligned}\frac{\partial \varphi}{\partial x}(x, i) &= ci^2 \cdot 2(\delta - x)^{-3} \cdot (-1) \\ &= \frac{2ci^2}{(\delta - x)^3}.\end{aligned}\quad (26)$$

Partial derivative with respect to i :

$$\frac{\partial \varphi}{\partial i}(x, i) = \frac{2ci}{(\delta - x)^2}.\quad (27)$$

The gradient vector is therefore:

$$\nabla \varphi(x, i) = \begin{bmatrix} \frac{2ci^2}{(\delta - x)^3} \\ \frac{2ci}{(\delta - x)^2} \end{bmatrix}.\quad (28)$$

5.3 Taylor Expansion of φ at the Operating Point

The first-order Taylor approximation about (x_0, i_0) is:

$$\begin{aligned}\varphi(x, i) &\approx \varphi(x_0, i_0) + \frac{\partial \varphi}{\partial x}(x_0, i_0)(x - x_0) \\ &\quad + \frac{\partial \varphi}{\partial i}(x_0, i_0)(i - i_0),\end{aligned}\quad (29)$$

where:

$$\varphi(x_0, i_0) = \frac{ci_0^2}{(\delta - x_0)^2},\quad (30)$$

$$\left. \frac{\partial \varphi}{\partial x} \right|_0 = \frac{2ci_0^2}{(\delta - x_0)^3},\quad (31)$$

$$\left. \frac{\partial \varphi}{\partial i} \right|_0 = \frac{2ci_0}{(\delta - x_0)^2}.\quad (32)$$

5.4 Linearised Jacobian Entries

Defining deviation variables $\Delta x = x - x_0$, $\Delta v = \dot{x} - v_0 = \dot{x}$, $\Delta i = i - i_0$, and noting $\frac{\partial}{\partial x}[-k(x - d)] = -k$ and $\frac{\partial}{\partial v}[-bv] = -b$, the Jacobian entries of $\dot{v} = f(x, v, i)$ evaluated at the operating point are:

$$\left. \frac{\partial f}{\partial x} \right|_0 = \frac{5}{7m} \left(-k + \frac{2ci_0^2}{(\delta - x_0)^3} \right),\quad (33)$$

$$\left. \frac{\partial f}{\partial v} \right|_0 = \frac{5}{7m}(-b),\quad (34)$$

$$\left. \frac{\partial f}{\partial i} \right|_0 = \frac{5}{7m} \cdot \frac{2ci_0}{(\delta - x_0)^2}.\quad (35)$$

5.5 Linearised State Equations

The linearised system about the equilibrium $(x_0, 0, i_0)$ is:

$$\Delta \dot{x} = \Delta v,\quad (36)$$

$$\Delta \dot{v} = \left. \frac{\partial f}{\partial x} \right|_0 \Delta x + \left. \frac{\partial f}{\partial v} \right|_0 \Delta v + \left. \frac{\partial f}{\partial i} \right|_0 \Delta i,\quad (37)$$

which can be written in state-space form as $\Delta \dot{\mathbf{x}} = A \Delta \mathbf{x} + B \Delta i$, where:

$$A = \begin{bmatrix} 0 & 1 \\ \left. \frac{\partial f}{\partial x} \right|_0 & \left. \frac{\partial f}{\partial v} \right|_0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \left. \frac{\partial f}{\partial i} \right|_0 \end{bmatrix}.\quad (38)$$

5.6 Full Three-State Linearised Model

Including the electrical state $x_3 = i$ (current), the complete linearised state vector is $\bar{\mathbf{x}} = [\Delta x_1, \Delta x_2, \Delta x_3]^\top$. Evaluating all partial derivatives numerically at $x_0 = 0.50$ m and $i_0 \approx 22.1$ A, the full state-space matrices are:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 133.0 & -16.08 & 20.68 \\ 0 & 0 & -15158.82 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 6.89 \end{bmatrix},\quad (39)$$

$$C = [1 \quad 0 \quad 0], \quad D = 0.\quad (40)$$

The input is the voltage deviation $\bar{u} = \Delta V$ and the output is the position deviation $\bar{y} = \Delta x_1$. The positive entry $a_{21} = 133.0$ reflects the destabilising magnetic force gradient, and the fast electrical pole $a_{33} = -15158.82$ confirms the time-scale separation between electrical and mechanical dynamics.

6 Electrical, Sensor, and Transfer Function Modelling

The previous sections derived and linearised the mechanical dynamics of the system. However, this model does not yet reflect the physical control input, which is the applied voltage V . The actuator and sensor dynamics must be included before deriving the transfer function used for control design.

6.1 Electrical Dynamics

The inductance of the electromagnet depends on the position of the ball:

$$L(x) = L_0 + L_1 e^{-\alpha(\delta - x)}.\quad (41)$$

Kirchhoff's voltage law gives:

$$V = Ri + \frac{d}{dt}(L(x)i).\quad (42)$$

Expanding this using the product rule,

$$\frac{d}{dt}(L(x)i) = L(x)\dot{i} + i\dot{L}(x).\quad (43)$$

Since $\dot{L}(x) = \frac{dL}{dx}\dot{x}$, the term $i\dot{L}(x)$ is proportional to the ball velocity. Near the operating point, $\dot{x} \approx 0$, so this term is small relative to $L(x)\dot{i}$ and is neglected in the first-order model. This gives:

$$\dot{i} = \frac{V - Ri}{L(x)}.\quad (44)$$

This shows that the electrical dynamics are coupled to the mechanical position through the position-dependent inductance.

6.2 Sensor Model

The measured position is obtained from a first-order sensor model:

$$\tau_m \dot{y}_m + y_m = x, \quad (45)$$

where $\tau_m = 0.03$ s.

Taking the Laplace transform,

$$H(s) = \frac{Y_m(s)}{X(s)} = \frac{1}{0.03s + 1}. \quad (46)$$

6.3 Transfer Function Derivation

The linearised system is written in standard state-space form as

$$\dot{\bar{x}} = A\bar{x} + B\bar{u}, \quad \bar{y} = C\bar{x} + D\bar{u}, \quad (47)$$

where $\bar{x}$, $\bar{u}$, and $\bar{y}$ are all expressed in deviation variables about the operating point.

Taking the Laplace transform with zero initial conditions [1] gives $\bar{X}(s) = (sI - A)^{-1}B\bar{U}(s)$, so that

$$G(s) = \frac{\bar{Y}(s)}{\bar{U}(s)} = C(sI - A)^{-1}B + D. \quad (48)$$

Substituting the matrices from (39)–(40) gives

$$G(s) = \frac{142.5}{s^3 + 15170s^2 + 243600s - 2016000}. \quad (49)$$

Including the sensor dynamics $H(s) = 1/(0.03s + 1)$, The overall measured transfer function is

$$G_{\text{total}}(s) = \frac{142.5}{D(s)}, \quad (50)$$

where

$$D(s) = (s^3 + 15170s^2 + 243600s - 2016000)(0.03s + 1). \quad (51)$$

6.4 Open-Loop Stability

The poles of $G(s)$ are the roots of its denominator:

$$s_1 \approx -15158.82, \quad s_2 \approx -22.10, \quad s_3 \approx 6.02. \quad (52)$$

The sensor introduces an additional pole $s_4 = -33.33$.

The right-half-plane pole at $s_3 \approx +6.02$ rad/s confirms open-loop instability [1]: as the ball moves toward the electromagnet, the attractive force increases as $1/(\delta - x)^2$, amplifying the deviation rather than restoring equilibrium. Feedback control is therefore essential to stabilise the system.

7 Controller Design and Closed-Loop Assessment

7.1 PID Controller Formulation

The controller was designed using the linearised model about the feasible operating point $x_0 = 0.50$ m. In deviation variables, the PID control law is

$$u_{\text{dev}}(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \dot{e}(t), \quad (53)$$

where $e(t) = r(t) - y(t)$ is the tracking error [1]. In the Laplace domain,

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}. \quad (54)$$

The open-loop, closed-loop, and sensitivity transfer functions are:

$$L(s) = C(s)G(s)H(s), \quad (55)$$

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)}, \quad (56)$$

$$S(s) = \frac{1}{1 + C(s)G(s)H(s)}. \quad (57)$$

The integral term makes the controller type-1, ensuring zero steady-state error and rejection of constant disturbances for a stable closed loop.

7.2 Controller Input in Deviation Variables

Because the system is linearised about a non-zero equilibrium, the PID output is a deviation voltage rather than the total physical input. The actual voltage applied to the circuit is

$$V(t) = V_0 + u_{\text{dev}}(t), \quad (58)$$

where $V_0 = Ri_0$ is the equilibrium bias voltage. For $x_0 = 0.50$ m:

$$V_0 \approx 2200 \times 22.1 \approx 4.86 \times 10^4 \text{ V}. \quad (59)$$

This large value arises directly from the circuit resistance $R = 2.2 \text{ k}\Omega$ and represents a practical limitation of the actuator.

7.3 Candidate PID Gains

The Routh–Hurwitz stability criterion [1] was applied to the closed-loop characteristic polynomial to derive necessary conditions on the PID gains before any simulation was carried out. The characteristic polynomial of the closed-loop system is fifth-order:

$$Q(s) = a_0 s^5 + a_1 s^4 + a_2 s^3 + a_3 s^2 + a_4 s + a_5, \quad (60)$$

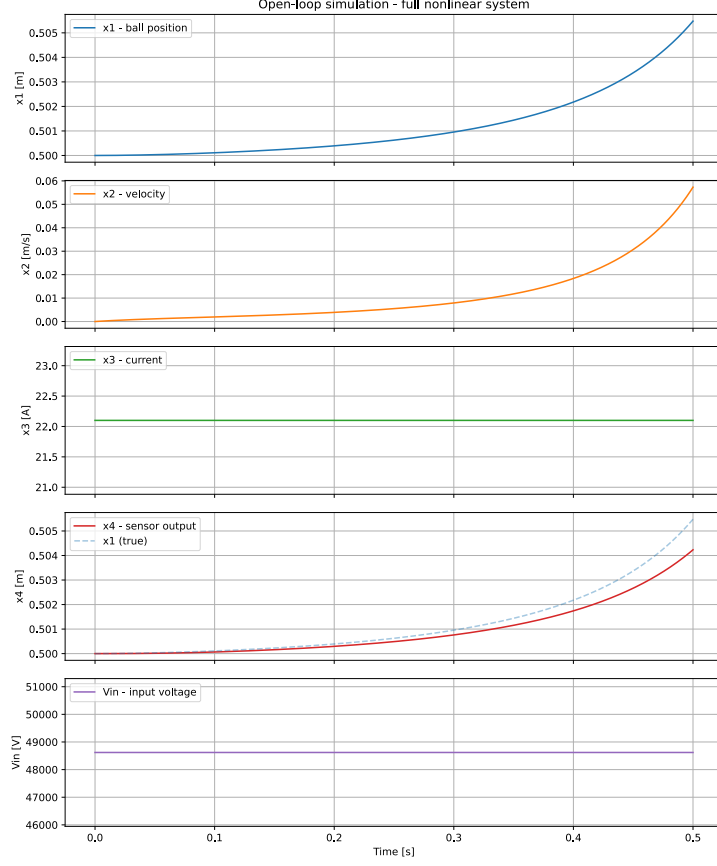


Figure 1: Open-loop simulation showing exponential divergence of ball position from equilibrium $x_0 = 0.50$ m, confirming the RHP pole at $s \approx +6.02$ rad/s.

where the coefficients depend on the gains as follows:

$$a_3 = 142.5K_d + 183120, \quad (61)$$

$$a_4 = 142.5K_p - 2,016,000, \quad (62)$$

$$a_5 = 142.5K_i. \quad (63)$$

For all first-column entries of the Routh table to be strictly positive, the following necessary conditions must hold:

$$K_i > 0, \quad (64)$$

$$K_p > \frac{2,016,000}{142.5} \approx 14,147, \quad (65)$$

$$K_d > -1285 \quad (\text{at } K_p \approx 14,147). \quad (66)$$

The condition (65) arises directly from the right-half-plane open-loop pole at $s \approx +6.02$, which introduces the negative coefficient $-2,016,000$ into the s^1 term of $Q(s)$. These three conditions are necessary but not sufficient: a fifth condition on the s^1 row element d_1 couples all three gains and was verified numerically for each candidate design.

Gains near the minimum $K_p \approx 14,147$ place the dominant closed-loop pole very close to the imaginary axis, giving settling times exceeding 30 s. Gains sufficiently above the minimum to achieve acceptable settling unavoidably produce significant overshoot due to the excited complex pole pair. This

represents a fundamental limitation of PID control applied to this open-loop unstable plant, rather than a tuning deficiency. Three candidate gain sets were evaluated using closed-loop pole analysis and step-response simulation, with results summarised in Table 1.

7.4 Selected Controller

The final controller selected was

$$K_p = 32000, \quad K_i = 20000, \quad K_d = 1500, \quad (67)$$

giving

$$\begin{aligned} C(s) &= 32000 + \frac{20000}{s} + 1500s \\ &= \frac{1500s^2 + 32000s + 20000}{s}. \end{aligned} \quad (68)$$

This gave the best overall compromise: BIBO stability, essentially zero steady-state error, and the lowest peak control deviation of the improved designs.

Table 1: Summary of the PID gain sets considered.

Gains (K_p, K_i, K_d)	OS (%)	T_s (s)	Peak $ u_{dev} $ (V)	Comment
(20000, 5000, 50)	≈ 248	≈ 4.9	$\approx 5.25 \times 10^4$	Stable, but the overshoot is far too large for an acceptable final design.
(40000, 40000, 1500)	≈ 75.7	≈ 2.0	$\approx 3.93 \times 10^4$	Faster response, but still associated with high overshoot and a larger control demand.
(32000, 20000, 1500)	≈ 80.7	≈ 3.0	$\approx 3.12 \times 10^4$	Chosen as the best overall compromise among the candidate PID designs tested.

Table 2: Stability margins for the selected controller.

Quantity	Value	Frequency (rad/s)
Phase margin (PM)	41.4°	$\omega_{gc} = 11.20$
Gain margin (GM)	-6.64 dB	$\omega_{pc} = 2.17$

7.5 Closed-Loop Stability Verification

For the selected gains, the closed-loop characteristic polynomial is

$$Q(s) = 0.03s^5 + 456.1s^4 + 22478s^3 + 396870s^2 + 2.544 \times 10^6 s + 2.85 \times 10^6. \quad (69)$$

Applying the Routh criterion gives the first-column entries

$$[0.03, 456.1, 22451.9, 345194, 2.358 \times 10^6, 2.85 \times 10^6], \quad (70)$$

which are all strictly positive, confirming no right-half-plane closed-loop poles. Direct root calculation gives:

$$\begin{aligned} s_1 &\approx -15153.95, & s_2 &\approx -11.81, \\ s_{3,4} &\approx -18.08 \pm 7.14j, & s_5 &\approx -1.40. \end{aligned} \quad (71)$$

Since all poles lie in the open left half-plane, the linearised closed-loop system is BIBO stable [1].

7.6 Stability Margins

The gain and phase margins were computed from the open-loop Bode plot (Fig. 3) and are summarised in Table 2.

The PM of 41.4° is just below the commonly used 45° guideline, indicating adequate but not generous robustness. The negative GM is unreliable here because the Bode criterion assumes open-loop stability, which does not hold due to the RHP pole, and the phase crosses -180° twice. Stability is confirmed by the Routh criterion (70).

7.7 Closed-Loop Performance

The selected controller produces a final output value very close to the reference, confirming negligible steady-state error from the integral action. The step

response returns smoothly to the setpoint without sustained oscillation.

The main weakness is the large overshoot ($\approx 81\%$) and large control effort. As established from the Routh analysis, this is a structural consequence of the RHP open-loop pole forcing $K_p > 14,147$, not a tuning deficiency. The chosen gains represent the best compromise obtainable with a PID structure on this unstable plant.

7.8 Practical Limitations

The controller was designed on the linearised model, and the validity of the linear approximation decreases as the state moves away from the operating point. Anti-windup protection and actuator saturation limits would be required in any real implementation given the large bias voltage $V_0 \approx 4.86 \times 10^4$ V.

7.9 Nonlinear Validation

To assess whether the controller works on the real nonlinear system, the closed-loop simulation was run using the full nonlinear equations of motion. The bias voltage was recomputed at the setpoint using $V_0 = Ri_{sp}$ to avoid integrator windup from a persistent bias mismatch.

Fig. 4 shows the result for a step to $x_{sp} = 0.48$ m. The nonlinear response (solid) and linear prediction (dashed) both converge to the setpoint, confirming that the controller is effective for small setpoint deviations close to $x_0 = 0.50$ m.

Testing at setpoints above x_0 revealed a practical operating boundary. For $x_{sp} \geq 0.49$ m, the transient overshoot carries the ball close enough to the electromagnet that the nonlinear magnetic force overwhelms the controller before it can recover. The effective region of attraction is therefore asymmetric: downhill setpoints ($x_{sp} < x_0$) are handled reliably, while uphill setpoints are limited. A comprehensive solution would require re-linearisation at each new setpoint or a nonlinear control strategy.

7.10 Disturbance Rejection

The type-1 controller guarantees zero steady-state error to constant disturbances. To verify this on the nonlinear system, a voltage disturbance of 1500 V

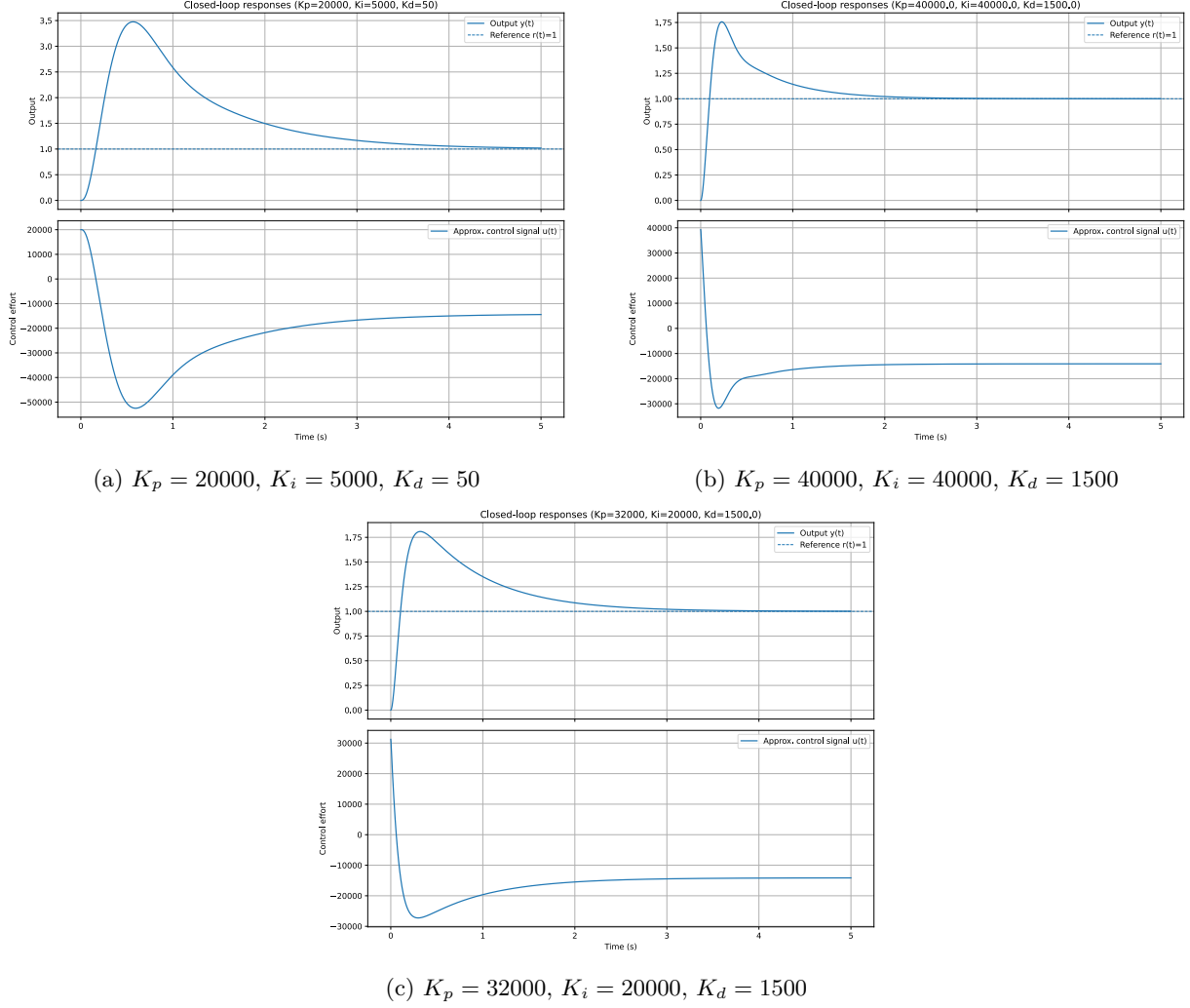


Figure 2: Closed-loop responses of the three PID designs considered. Each plot shows the output response and the corresponding control deviation $u_{\text{dev}}(t)$. The initial trial controller is clearly unacceptable, whereas the latter two provide a more reasonable balance between response speed and damping.

was applied for 0.2 s at $t = 3$ s. Fig. 5 shows a brief deviation of ≈ 0.018 m, after which integral action drives the ball back to x_{sp} with zero steady-state error, satisfying requirement (iv).

7.11 Summary of the Control Design

A PID controller was selected because it is fully supported by the module syllabus and provides a reasonable baseline for the linearised plant. The selected gains yield a BIBO-stable closed loop with essentially zero steady-state error and verified disturbance rejection. The primary engineering trade-off is between settling time and overshoot, which is unavoidable given the structural constraint imposed by the RHP open-loop pole. Practical implementation would require anti-windup protection and consideration of actuator voltage limits.

8 Conclusion

This report presented the full modelling, linearisation, and PID control design for a ball rolling on an inclined plane under electromagnetic actuation. The nominal setpoint $x_{\text{sp}} = 0.40$ m was shown to be physically infeasible; the system was linearised about the feasible operating point $x_0 = 0.50$ m instead. The open-loop plant is unstable due to a right-half-plane pole at $s \approx +6.02$ rad/s, which imposes the hard gain constraint $K_p > 14,147$ and makes significant overshoot unavoidable with a PID structure. The selected gains $(K_p, K_i, K_d) = (32000, 20000, 1500)$ produce a BIBO-stable closed loop with negligible steady-state error, a phase margin of 41.4° , and verified disturbance rejection on the full nonlinear model. The primary practical limitations are the large equilibrium bias voltage $V_0 \approx 4.86 \times 10^4$ V arising from $R = 2.2$ k Ω , and the small asymmetric region of attraction identified in the nonlinear validation.

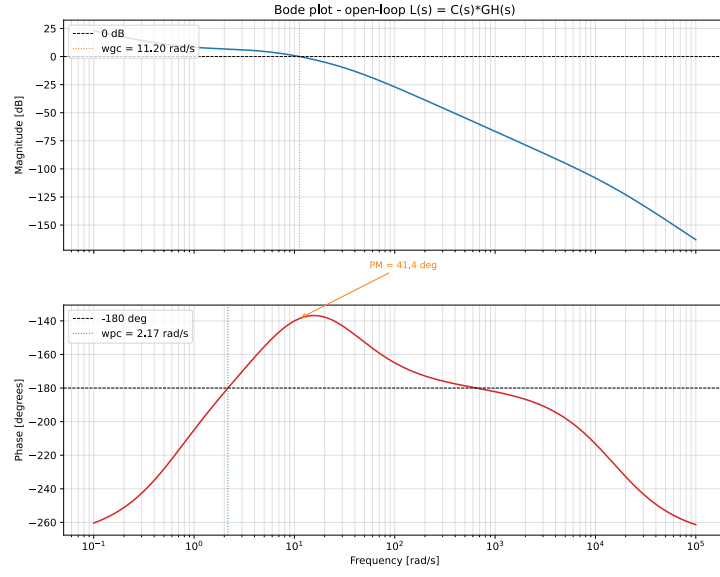


Figure 3: Bode plot of $L(s) = C(s)G(s)H(s)$. $PM = 41.4^\circ$ at $\omega_{gc} = 11.20$ rad/s. The phase crosses -180° twice due to the RHP pole, making the GM unreliable.

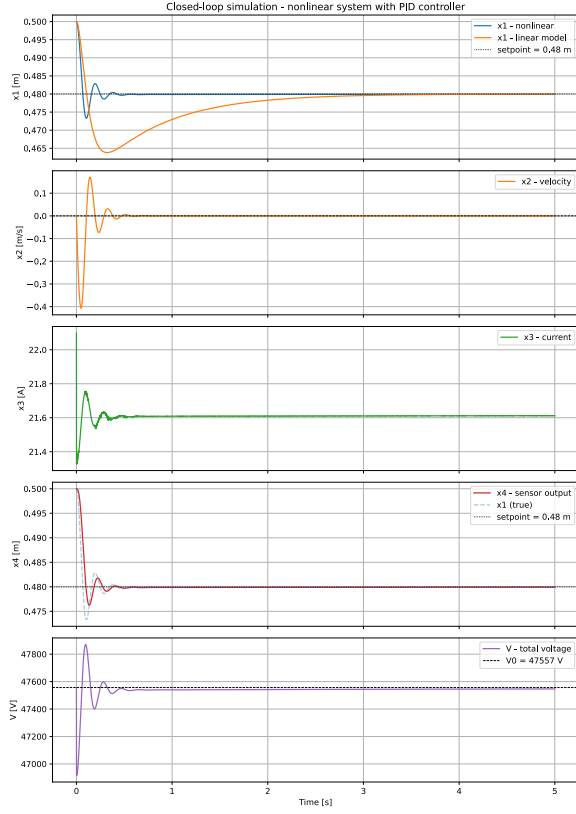


Figure 4. Closed-loop nonlinear simulation for a step to $x_{sp} = 0.48$ m. Both the nonlinear (solid) and linearised (dashed) responses converge to the setpoint, confirming closed-loop stability on the nonlinear plant.

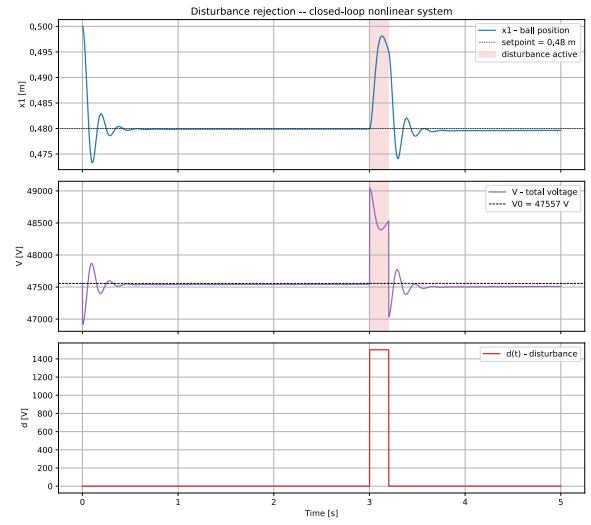


Figure 5. Disturbance rejection simulation. A 1500 V pulse at $t = 3$ s (shaded) causes a brief deviation before integral action restores zero steady-state error.

References

- [1] P. Sopasakis, *Control Systems: An Introduction*, v1.0.5-rc.9 ed. Belfast, UK: Applied Mathematix Press, 2023.

Appendix: Collaboration Report

A.1 Meeting Minutes

Table 3: Summary of meeting minutes

Date	Attendees	Progress Review	Conclusions / Actions
21/01/2026	DM, JM, CC	Kick-off meeting. Reviewed assignment brief.	A1.1: DM to set up Overleaf document. A1.2: CC to create GitHub repository and invite all members. A1.3: JM to draft Gantt chart. A1.4: All to read assignment and identify tasks for next meeting.
28/01/2026	DM, JM, CC	A1.1, A1.2 completed. A1.3 in progress. Began nonlinear model derivation.	A2.1: DM to derive mechanical equations of motion. A2.2: JM to derive electrical model and sensor dynamics. A2.3: CC to verify equilibrium conditions numerically in Python.
16/02/2026	DM, JM, CC	Mechanical and electrical models complete. Equilibrium infeasibility at $x_0 = 0.40$ m identified and resolved.	A3.1: JM to complete linearisation and Jacobian derivation. A3.2: CC to implement state-space model in Python and verify poles. A3.3: JM to begin transfer function derivation.
24/02/2026	DM, JM, CC	Linearisation and transfer function complete. Python simulation running. Open-loop instability confirmed.	A4.1: All to investigate PID gain tuning. A4.2: DM to run Routh analysis for candidate gains. A4.3: CC to produce step response plots. A4.4: DM to write controller design section.
30/03/2026	DM, JM, CC	PID gains selected. Stability verified via Routh and pole calculation. Report draft assembled.	A5.1: All to review and edit full report draft. A5.2: CC to add disturbance rejection and nonlinear validation figures. A5.3: DM to finalise collaboration appendix. A5.4: JM to proofread equations and references.

A.2 Collaboration Technologies

The team used the following tools throughout the project:

- **Overleaf** — collaborative report writing, with all three members contributing to the shared document simultaneously.
- **Python** — all modelling, simulation, and control design was implemented in Python, with code shared and version-controlled via the GitHub repository available at: <https://github.com/ccogh/ELE2038-Signals-and-Control-Group-Coursework---Group-/tree/main>.
- **GitHub** — version control for the Python code, allowing all members to track changes and contributions.
- **WhatsApp** — used for day-to-day communication, sharing quick updates, and coordinating availability between meetings.
- **Microsoft Teams** — used for online meetings, allowing the team to discuss progress, review derivations, and agree on actions when meeting in person was not possible.

A.3 Task Management

Tasks were distributed at each meeting and recorded in the minutes above. Each team member was assigned specific sections and actions, and progress was reviewed at the start of each subsequent meeting. Outstanding actions were tracked informally via the GitHub repository issue tracker, allowing all members to see the current state of open tasks between meetings.

A.4 Individual Contributions

Member	Contribution
Dara McCollam	Derived the mechanical equations of motion including Newton's second law, rolling constraint, and moment of inertia for a solid sphere. Performed the equilibrium analysis and identified the infeasibility of $x_0 = 0.40$ m. Conducted Routh stability analysis for all candidate gain sets and derived the minimum gain condition $K_p > 14,147$. Designed and tuned the PID controller. Structured and set up the L ^A T _E X report template on Overleaf.
Joseph McNeill	Derived the electrical circuit model including Kirchhoff's voltage law, position-dependent inductance, and the justification for neglecting the $i\dot{L}$ term. Derived the sensor transfer function $H(s)$. Performed the linearisation of the full 3-state system and computed the Jacobian entries symbolically and numerically. Computed the plant transfer function $G(s) = C(sI - A)^{-1}B$ and identified the open-loop poles, confirming instability due to the right-half-plane pole at $s \approx 6.02$.
Ciaran Coghlan	Created and managed the GitHub repository, maintaining version control for all Python code. Implemented the nonlinear closed-loop simulation in Python and produced the nonlinear validation figure comparing the linearised and nonlinear step responses. Produced the disturbance rejection simulation and figure, demonstrating requirement (iv) of the assignment. Contributed to the Python implementation of the state-space model and step-response plots.

A.5 Additional Information

What went well:

- Task distribution was clear from the first meeting, with each member taking ownership of a specific part of the modelling pipeline.
- Use of GitHub allowed all members to track changes to the Python code and resolve discrepancies in numerical results efficiently.
- Regular meetings meant that errors (such as the infeasibility of the nominal setpoint) were caught early and corrected before affecting downstream work.

Challenges faced:

- Reconciling the sign conventions between the mechanical and electrical sections required an extra meeting to align notation.
- The large equilibrium voltage ($V_0 \approx 4.86 \times 10^4$ V) was surprising and required careful verification before the team was confident in the model. The cause was identified as the large circuit resistance $R = 2.2 \text{ k}\Omega$, which directly multiplies the equilibrium current in $V_0 = Ri_0$.